Fusion Data Card

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Card ID

Primary Dataset	*“On_Time_Reporting_Carrier_On_Time_Performance_(1987_present)_2026_6.csv”*
Description	*This dataset contains 607,577 US domestic flight records reported by 12 airlines across 351 airports, with scheduled and actual times, delays by cause, cancellations and diversions, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 1	*“nldn-tiles-202606.csv”*
Description	*This dataset contains 475,343 daily lightning-strike counts on a 0.1° latitude/longitude grid covering the continental US, totalling 4,538,490 strikes, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 2	*“hail-tiles-202606.csv”*

Description	*This dataset contains 338,241 daily radar-detected hail counts on the same 0.1° grid, totalling 2,428,076 detections, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 3	*“airports.csv”*
Description	*This dataset contains 86,135 airports, heliports and airfields worldwide from OurAirports, with their latitude, longitude and IATA code. We use it only to give each BTS airport a location, because the flight data has no coordinates.*

Datasets at a glance

Primary dataset

The primary dataset contains 607,577 examples representing individual US domestic flight segments (One row per flight) in the period from 06/01/2026 to 06/30/2026. It covers 12 reporting carriers and 351 origin and destination airports. In this period 1.65% of flights were cancelled and 0.39% diverted, with a mean departure delay of 20.7 minutes.

Fields (if applicable)

Field name	Data Type	Description
FlightDate	date (string)	Flight date, YYYY-MM-DD. 2026-06-01 to 2026-06-30. e.g. 2026-06-16
Reporting_Airline	string	Carrier code. 12 values: AA, AS, B6, DL, F9, G4, MQ, OH, OO, UA, WN, YX
Origin / Dest	string	Origin/destination airport code. 351 distinct. e.g. MKE, DTW
CRSDepTime	integer	Scheduled departure, local hhmm. 1–2359. e.g. 1649

Field name	Data Type	Description
DepTime	integer	Actual departure, local hhmm. 1–2400. Null if cancelled
DepDelayMinutes	float	Departure delay, minutes; negative values floored to 0. 0–2759
ArrDelayMinutes	float	Arrival delay, minutes. 0–2752
Cancelled	float (0/1)	1 if cancelled. 10,019 flights (1.65%)
CancellationCode	string	Reason: A=Carrier, B=Weather, C=National Air System, D=Security
Diverted	float (0/1)	1 if diverted. 2,363 flights (0.39%)
WeatherDelay	float	Minutes attributed to weather. 0–1823
NASDelay	float	Minutes attributed to the National Air System. 0–1310
CarrierDelay	float	Minutes attributed to the carrier. 0–2692
LateAircraftDelay	float	Minutes from a late inbound aircraft. 0–2398
Distance	float	Great-circle distance, miles. 31–4983

Supplemental dataset 1

The supplemental dataset contains 475,343 examples representing one grid-cell-day of lightning activity each, across 129,733 distinct 0.1-degree cells spanning longitude −125.6 to −44.5 and latitude 16.6 to 51.7, in the period from 06/01/2026 to 06/30/2026.

Fields (if applicable)

Field name	Data Type	Description
ZDAY	date (integer)	UTC day, YYYYMMDD. 20260601–20260630. e.g. 20260601
CENTERLON	float	Grid cell centre longitude, degrees. −125.6 to −44.5. e.g. -96.0
CENTERLAT	float	Grid cell centre latitude, degrees. 16.6 to 51.7. e.g. 37.6
TOTAL_COUNT	integer	Lightning strikes in that cell that day. 1–1,481, mean 9.5

Supplemental dataset 2

The supplemental dataset contains 338,241 examples representing one grid-cell-day of radar-detected hail each, across 93,653 cells spanning longitude −167.8 to 147.9 and latitude 10.5 to 67.4, in the period from 06/01/2026 to 06/30/2026.

Fields (if applicable)

Field name	Data Type	Description
ZDAY	date (integer)	UTC day, YYYYMMDD. 20260601–20260630
CENTERLON	float	Grid cell centre longitude, degrees. −167.8 to 147.9. e.g. -79.9
CENTERLAT	float	Grid cell centre latitude, degrees. 10.5 to 67.4. e.g. 33.0
TOTAL_COUNT	integer	Hail detections in that cell that day. 1–228, mean 7.2

Supplemental dataset 3

The supplemental dataset contains 86,135 examples, one per airport, heliport, seaplane base or closed

airfield worldwide. Only 9,054 of them have an IATA code, and 351 of those match the BTS origin and destination airports (97 large, 252 medium and 2 small airports). Added in Milestone 1, because the join to the weather grid needs a location for each airport and the BTS data has none.

Fields (if applicable)

Only the fields we use are listed. The file has 19 columns.

Field name	Data Type	Description
iata_code	string	Three-letter IATA code, the same code BTS uses for Origin/Dest. Empty for 77,081 rows. e.g. AUS
name	string	Airport name. e.g. Austin Bergstrom International Airport
type	string	large_airport, medium_airport, small_airport, heliport, seaplane_base, balloonport or closed
latitude_deg	float	Airport latitude, degrees. e.g. 30.1975
longitude_deg	float	Airport longitude, degrees. e.g. -97.6620
iso_country	string	Country code. e.g. US

Additional notes

- hail-tiles-202606.csv misdescribes itself in its own header comment as “the count of Vaisala NLDN lightning strikes” — identical boilerplate to the lightning file. This is a copy-paste error in NOAA’s header generation, present across their hail, structure and lightning tile products. The data are hail detections; only the column names and filename are reliable.
- The five delay-cause fields are populated for only 153,037 of 607,577 rows, because BTS attributes causes only when arrival delay reaches 15 minutes. These nulls are not missing at random, and treating them as zero is a modelling decision rather than a cleaning step.
- The primary CSV’s header row ends in a trailing comma, producing an empty 110th column that pandas names Unnamed: 110.
- Spatial extents differ between the supplemental sets: lightning covers the continental US, while hail extends to Alaska, Pacific and Caribbean radar sites and crosses the antimeridian. National maps

need consistent extents.

- OurAirports lists Palm Beach International under its new code DJT, but the June 2026 BTS file still uses PBI. We map DJT to PBI in the notebook.

Usage information

Questions driving the exploration study

1. How does the daily volume of lightning near an airport relate to that airport's mean departure delay?
2. Which of the 351 airports show the strongest weather-delay sensitivity, and does that track their geography?
3. Does BTS's own WeatherDelay attribution agree with independently observed lightning and hail activity on the same date and location?
4. How do the five delay causes shift in composition on days with severe weather versus clear days?
5. Do hail days and lightning days produce different operational responses (Cancellations versus delays)?
6. How far does weather disruption propagate through LateAircraftDelay to airports with no local severe weather?
7. Which carriers absorb weather disruption best, controlling for their route geography?

Sanctioned Use(s)

- BTS On-Time Performance: collected under 14 CFR Part 234 for public reporting of airline service quality, and published to support the DOT's Air Travel Consumer Report. Public analysis and research are explicitly sanctioned uses.
- NOAA SWDI (both supplemental sets): NCEI publishes the Severe Weather Data Inventory as a research and analysis archive of severe-weather records.
- OurAirports: a community-maintained open dataset, published for any use.
- Discouraged: SWDI hail, mesocyclone and tornado-signature products are radar-algorithm output, not verified ground reports — the raw point files label themselves #This file contains experimental data. They should not be treated as confirmed hailfall, nor substituted for NCEI's Storm Events Database where ground truth matters. Coverage is limited to NEXRAD radar range.

Licensing

- BTS: a work of the US federal government, in the public domain under 17 U.S.C. §105. No restrictions on sharing, transformation or reuse. BTS requests citation as the source.
- NOAA SWDI: NOAA open data — free to use and redistribute, with attribution requested for unaltered data.
- The lightning data originates from Vaisala's commercially-operated NLDN. Under NOAA's contract, raw flash and strike-level data are restricted to government and military users. The 0.1-degree daily tile aggregate we use is the publicly releasable derivative — that aggregation is why the tiles product exists. Our file is unrestricted, but strike-level detail cannot be reconstructed from it.
- OurAirports: released into the public domain by its maintainers. No restrictions and no attribution required.

Privacy considerations

No personally identifiable information in any of the four datasets. The BTS data is operational and flight-level: no passenger, crew or booking records. Tail_Number identifies a specific airframe and no anonymisation or redaction was applied or required. The two SWDI datasets are geophysical observations aggregated to a spatial grid, with no human subjects. The OurAirports data describes public facilities only.

Format

- The primary dataset is in a CSV file.
- The first supplemental dataset is in a CSV file.
- The second supplemental dataset is in a CSV file.
- The third supplemental dataset is in a CSV file.
- The processed outputs of Milestone 1 are Parquet files in the bucket's processed/ folder: fused.parquet (one row per airport-day) and flights_clean.parquet (one row per flight).

Source and collection process

Origin

- Primary: Reporting Carrier On-Time Performance (1987–present) table, "On-Time" database, BTS TranStats data library: https://www.transtats.bts.gov/ontime/

- Supplemental 1 and 2: NOAA NCEI Severe Weather Data Inventory, CSV archive: https://www.ncei.noaa.gov/pub/data/swdi/database-csv/v2/ (files nldn-tiles-202606.csv.gz and hail-tiles-202606.csv.gz)
- Supplemental 3: OurAirports open data, https://ourairports.com/data/ (file airports.csv, from the mirror at https://davidmegginson.github.io/ourairports-data/airports.csv)

Original datasets author/publisher

- Primary: US Department of Transportation, Bureau of Transportation Statistics, Office of Airline Information. Collected under 14 CFR Part 234 from carriers accounting for at least 0.5% of domestic scheduled passenger revenues — which is why only 12 airlines appear.
- Supplemental: NOAA National Centers for Environmental Information (NCEI), Asheville, NC. Lightning observations are sourced from Vaisala Inc.'s National Lightning Detection Network under contract; hail detections derive from the NEXRAD Level-III Hail Detection Algorithm.
- Supplemental 3: OurAirports, a volunteer project started by David Megginson, with edits from its user community.

Original datasets point of contact

- Primary: BTS TranStats support — https://www.transtats.bts.gov/
- Supplemental: NCEI Customer Engagement Branch (ncei.orders@noaa.gov)
- Supplemental 3: OurAirports, https://ourairports.com/

Original metadata

- Primary: readme.html, shipped with the BTS extract; converted and stored at Datasets/Primary/README.md. Field definitions also at https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FGJ
- Supplemental: SWDI documentation at https://www.ncei.noaa.gov/pub/data/swdi/database-csv/v2/ and https://www.ncei.noaa.gov/products/severe-weather-data-inventory
- Supplemental 3: data dictionary at https://ourairports.com/help/data-dictionary.html

Collection process

- Primary: self-reported by airlines to DOT monthly under the Airline Service Quality Performance programme; released ~2 months after month end. June 2026 was published 12 August 2026. Details: https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations

- Supplemental: NLDN sensors detect cloud-to-ground flashes continuously; NEXRAD radars run the hail detection algorithm per volume scan. NCEI aggregates both to daily counts per 0.1-degree grid cell on UTC day boundaries.
- Supplemental 3: volunteers add and correct airport records by hand, and the maintainers export the full table as CSV every night.

Collection or access date

Primary dataset downloaded 31 August 2026. Supplemental datasets 1 and 2 downloaded 31 August 2026. Supplemental dataset 3 downloaded 28 September 2026.

Refresh date/schedule

- Primary: monthly, roughly two months in arrears. June 2026 released 12 August 2026; July expected mid-September. The file we hold is a fixed monthly extract and will not change.
- Supplemental: monthly files are posted early in the following month and revised — nldn-tiles-202606 was written 5 July 2026, and NCEI's directory shows August 2026 files updated as recently as 31 August 2026. Our copies are point-in-time snapshots taken 31 August 2026.
- Supplemental 3: updated nightly. Our copy is a snapshot taken 28 September 2026. Airport coordinates almost never change, but codes can: the PBI to DJT rename is an example.

Cleaning and processing

Missing values

We found missing values with isna().sum() on each DataFrame, then checked the cause of each group before we chose how to handle it. The notebook uses assert statements to test each cause against the data, so a wrong assumption stops the notebook.

Field(s)	Missing	Cause	Handling
carrier_delay, weather_delay, nas_delay, security_delay, late_aircraft_delay	454,540 (74.8%)	BTS gives delay causes only for flights that arrive 15+ minutes late. The cause fields are present on every such flight and	Filled with 0. A missing cause means that no delay minutes were attributed, not that the value is unknown.

Field(s)	Missing	Cause	Handling
		missing on every other flight.	
dep_delay, dep_time, dep_del15	9,554 (1.6%)	Every one of these flights was cancelled.	Kept as missing. A 0 would count a cancelled flight as on time and understate storm effects. The cancelled flag records the outcome.
arr_delay, arr_time, arr_del15	12,382 (2.0%)	Exactly the cancelled flights plus the diverted flights.	Kept as missing, for the same reason.
cancellation_code	597,558 (98.4%)	The flight was not cancelled.	Kept as missing. This is correct.
tail_number	832 (0.14%)	Not reported by the airline.	Kept. No step uses this field.
Div3–Div5 fields, Unnamed: 109	100%	Empty columns.	Dropped with the other unused columns.
lightning, hail	0		Nothing to do.
airports.iata_code	77,081 (89.5%)	Heliports, airstrips and closed fields have no IATA code.	Rows without a code are dropped, because they cannot match a BTS airport.
fused: mean_dep_delay, mean_arr_delay	11 airport-days	Small airports where every flight that day was cancelled, so there is no delay to average.	Kept as missing. cancel_rate is 1.0 on these rows.
fused: lightning_count, hail_count	8,283 / 8,297 before fill	An airport-day whose tile has no row in the weather file.	Filled with 0. The tile files list only cells with at least one detection, so no row means no detections.

We also converted types so that missing values are stored correctly: the 0/1 delay flags use the nullable boolean type, and the hhmm clock times use the nullable Int64 type.

Aggregation

The flights are one row per flight, and the weather is one row per 0.1° tile per day. We grouped the flights by origin airport and date with groupby(['flight_date', 'origin']), so that the target grain is **airport × day**, the finest grain that both sides share. This turned 607,577 flights into 10,306 airport-days.
For each airport-day we compute:

- n_flights: the number of scheduled departures.
- mean_dep_delay, mean_arr_delay: the mean delay in minutes. We use the mean so that busy and quiet airports are comparable. The median would hide the long storm tail that we want to study.
- cancel_rate, divert_rate: the fraction of flights cancelled or diverted.
- carrier_delay, weather_delay, nas_delay, security_delay, late_aircraft_delay: the total delay minutes for each cause, so that the size of a disruption stays visible.

We then filtered the airport-days to the **contiguous US**. We dropped 890 airport-days in Alaska, Hawaii, Puerto Rico, the US Virgin Islands and the Pacific territories, which leaves 9,416 airport-days at 320 airports. The lightning network (NLDN) is built for the lower 48 states, so a zero lightning count outside it would mean “not measured”, not “no lightning”.

We did not smooth the data before the join. The daily mean departure delay is noisy (8.1 to 33.4 minutes, standard deviation 7.2), and a 7-day rolling mean shows its trend clearly. But the day-to-day spikes follow the storms (the daily mean delay has a correlation of 0.68 with the daily lightning count at airport tiles), and those spikes are the signal we want to study. We use smoothing only for trend charts.

Dates: the weather ZDAY is a UTC date, and flight_date is a local date. A UTC day starts at 7pm CDT (8pm EDT, 5pm PDT) the evening before, so a storm late in the local evening counts toward the next day. We accept this at a daily grain: most storms happen in the local afternoon, before midnight UTC, and the tile counts are daily totals that we cannot shift. This mismatch can only weaken the link between storms and delays. It cannot create a false one.

Joins of input sources

The join has two steps:

1. **Airports to coordinates.** A left join of the airport-days to airports.csv on the airport code (origin = iata_code) adds a latitude and longitude to each airport. All 351 BTS airports match after we map DJT back to PBI. An assert stops the notebook if any airport has no coordinates.
2. **Airport-days to weather.** We round each airport’s latitude and longitude to one decimal place. This gives the center of the 0.1° tile that contains the airport (about 11 × 9 km, close to the ~5 mile lightning “ramp closure” radius). Then we left join lightning and hail on **(date, latitude, longitude)**, where flight_date matches ZDAY.

We use left joins so that every airport-day stays in the result, with or without weather. A missing tile row means no detections, so we fill it with 0 (see Missing values). An assert checks that the join adds no rows, which proves that the weather keys are unique.
The result, fused.parquet, has 9,416 airport-days at 320 airports. 1,133 airport-days (12.0%) have lightning in the airport tile, and 1,119 (11.9%) have hail. A first comparison shows that the join carries a real signal:

	Mean departure delay	Cancellation rate	Weather-delay minutes per airport-day
Lightning in the airport tile	26.8 min	4.1%	319.5
No lightning	16.6 min	1.7%	48.7

A known limit: one tile can miss a storm just outside it. Summing the 3 × 3 block of tiles around each airport is the next thing to try.

Exploration study

Descriptive statistics

Primary dataset (after cleaning, 607,577 flights, 31 fields). The delay-cause fields are after the fill with 0.

Field	Count	Missing	Mean	Std	Min	Median	Max
dep_delay (min)	598,023	9,554	17.62	62.55	−34	−1	2,759
arr_delay (min)	595,195	12,382	11.68	64.51	−73	−5	2,752
distance (miles)	607,577	0	854.03	611.82	31	693	4,983
carrier_delay (min)	607,577	0	5.76	38.46	0	0	2,692

Field	Count	Missing	Mean	Std	Min	Median	Max
weather_delay (min)	607,577	0	1.26	18.31	0	0	1,823
nas_delay (min)	607,577	0	3.80	20.23	0	0	1,310
security_delay (min)	607,577	0	0.02	1.35	0	0	539
late_aircraft_delay (min)	607,577	0	7.87	34.20	0	0	2,398

Field	Type	Distinct values	Most common	Notes
flight_date	datetime	30		2026-06-01 to 2026-06-30
airline	category	12	WN (120,933 flights)	
origin / dest	category	351 / 351	ORD (31,374 departures)	
dep_time_blk	category	19	0700-0759 (45,602)	
cancelled	bool	2	False	1.65% true (10,019)
diverted	bool	2	False	0.39% true (2,363)
dep_del15 / arr_del15	boolean	2	False	26.6% / 25.7% true
cancellation_code	category	4	B = Weather (6,603)	A = Carrier 2,124, C = NAS 1,286, D = Security 6
tail_number	string	5,654	N480HA (286)	832 missing

Delays are strongly right-skewed: the median departure is 1 minute early, but the mean is 17.6 minutes late because of a long tail of large delays. Weather is the most common reason for a cancellation (66% of

cancellations with a code), but weather causes only 1.26 minutes of delay per flight on average, much less than late aircraft (7.87) and the carrier (5.76).

Supplemental datasets 1 and 2 (weather tiles).

Field	Count	Missing	Mean	Std	Min	Median	Max	Total
lightning TOTAL_COUNT	475,343	0	9.55	22.69	1	3	1,481	4,538,490
hail TOTAL_COUNT	338,241	0	7.18	9.83	1	4	228	2,428,076

Both tile files cover 30 days and use the same 0.1° grid: every latitude and longitude is a multiple of 0.1, and no (day, latitude, longitude) key repeats.

Supplemental dataset 3 (airports). 86,135 rows: 42,746 small airports, 23,224 heliports, 13,548 closed fields, 4,106 medium airports, 1,273 seaplane bases, 1,175 large airports and 63 balloon ports. 9,054 rows have an IATA code, and the 351 that match BTS are 97 large, 252 medium and 2 small airports.

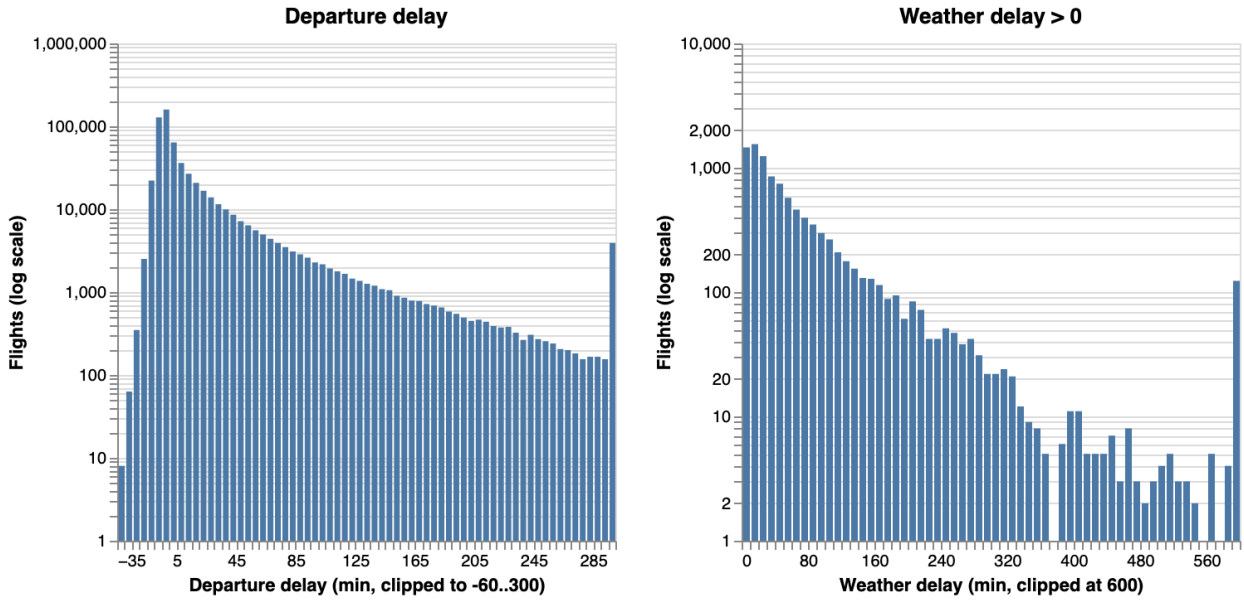
Processed dataset (fused, 9,416 airport-days at 320 airports).

Field	Count	Missing	Mean	Std	Min	Median	Max
n_flights	9,416	0	62.55	143.72	1	10	1,070
mean_dep_delay (min)	9,405	11	17.83	45.42	−23.5	8.56	1,197
mean_arr_delay (min)	9,405	11	12.71	47.25	−43	3.63	1,264
cancel_rate	9,416	0	0.02	0.07	0	0	1
divert_rate	9,416	0	0.00	0.03	0	0	0.5
weather_delay (total min)	9,416	0	81.28	550.98	0	0	22,064

Field	Count	Missing	Mean	Std	Min	Median	Max
nas_delay (total min)	9,416	0	242.93	765.65	0	17	17,207
late_aircraft_delay (total min)	9,416	0	500.31	1,669.25	0	33	47,765
lightning_count	9,416	0	1.53	10.31	0	0	307
hail_count	9,416	0	0.96	4.48	0	0	89

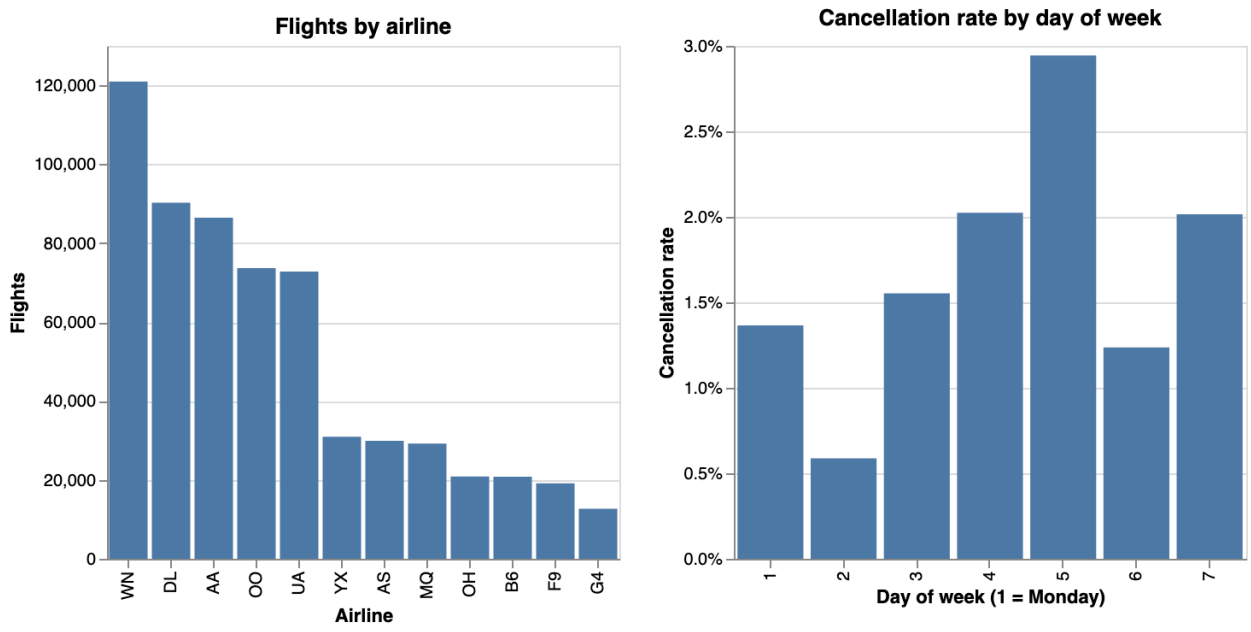
Airport size is very uneven: the median airport-day has 10 departures, but the largest has 1,070. Any comparison across airports must weight by flights or use rates.

Distribution plots



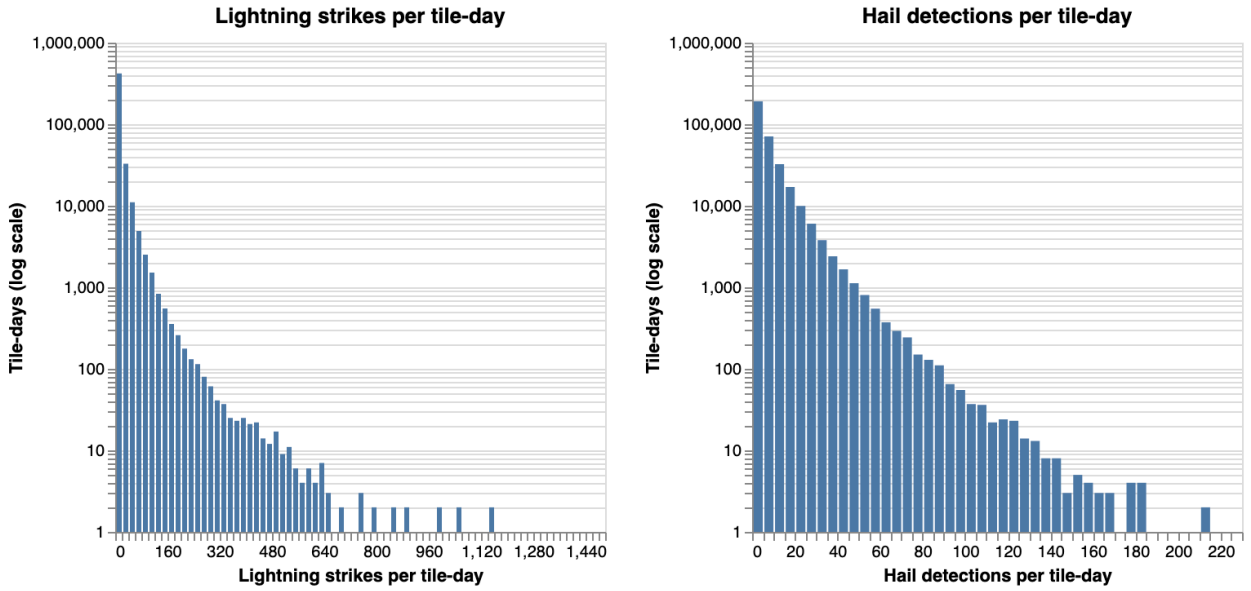
Departure delay (all flights) and weather delay (flights with weather delay > 0). Both use a log scale because of the long right tail.

The departure-delay distribution peaks just before 0 (most flights leave a few minutes early) and falls off slowly to the right. The spike at 300 is every flight delayed 300 minutes or more, collected by the clip. Weather delay is non-zero on 10,127 flights. Its median is 39 minutes and 64% of those delays are an hour or less, but 122 flights waited 10 hours or more.



Flights by airline, and cancellation rate by day of week.

Five airlines (WN, DL, AA, OO, UA) operate 73% of flights. The cancellation rate changes from 0.59% on Tuesdays to 2.94% on Fridays. June has only 4 or 5 of each weekday, so this probably shows which days had storms more than a real weekday effect.



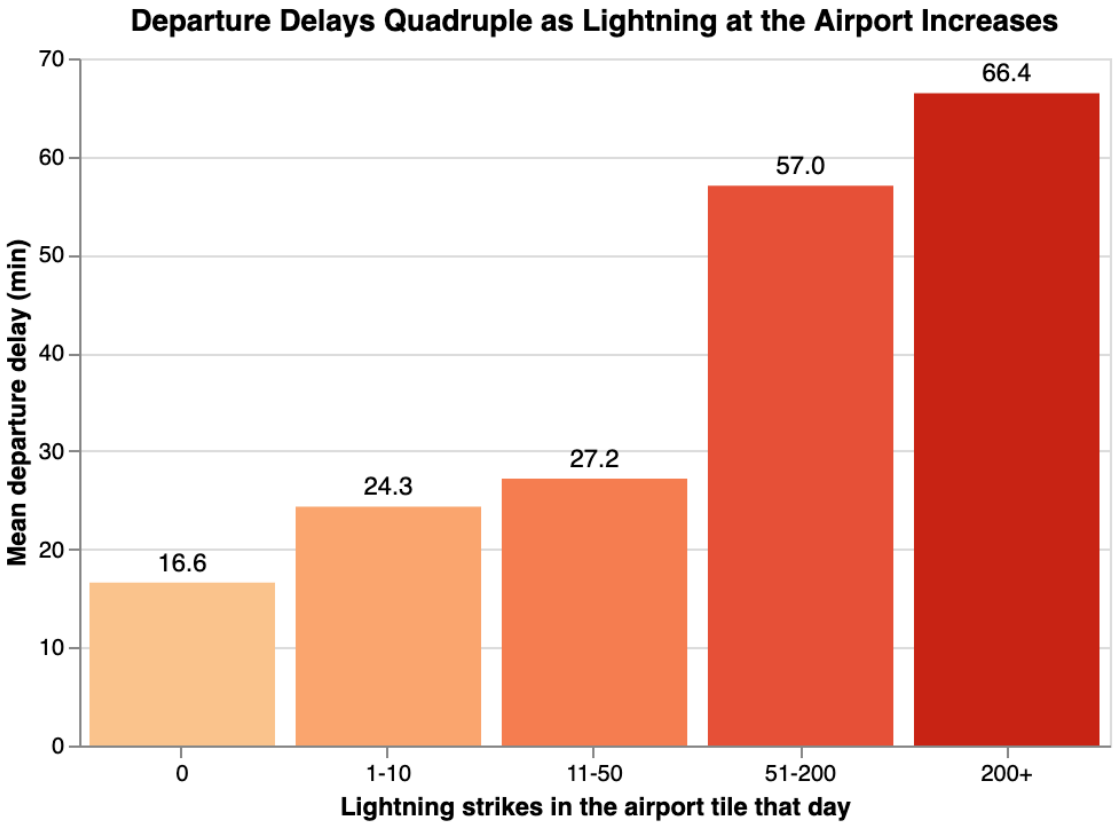
Lightning strikes and hail detections per tile-day, log scale.

Both weather counts are heavily skewed: most tile-days have only a few detections, and a small number of storm cells produce hundreds. This is why we compare “any lightning in the airport tile” against “none” before we use the raw counts.

Visual analysis

We asked seven questions, and the answers fall into three groups. We refer to each chart by its question number and the title of its refined version in milestone2/vis-analysis.ipynb.

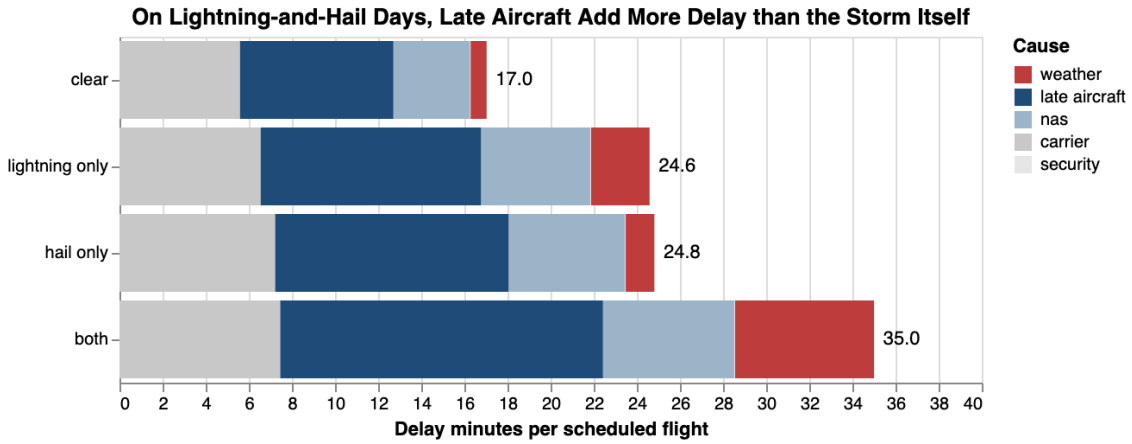
Storms at an airport slow it down, which we expected. The more useful part is how fast the delay grows. In Question 1, "Departure Delays Quadruple as Lightning at the Airport Increases", the mean departure delay is 16.6 minutes on days with no lightning in the airport's tile and 57.0 minutes once there are more than 50 strikes. Question 5, "Hail Alone and Lightning Alone Look Alike; Together They Cancel 6% of Flights", was more of a surprise. Hail days and lightning days are almost identical (about 2% of flights cancelled either way), but on days with both, airlines cancel about five times as many flights as on a clear day. How strong a storm is matters more than what kind it is, so our dashboard should measure storm strength rather than show a simple yes/no storm flag.



Question 1: "Departure Delays Quadruple as Lightning at the Airport Increases"

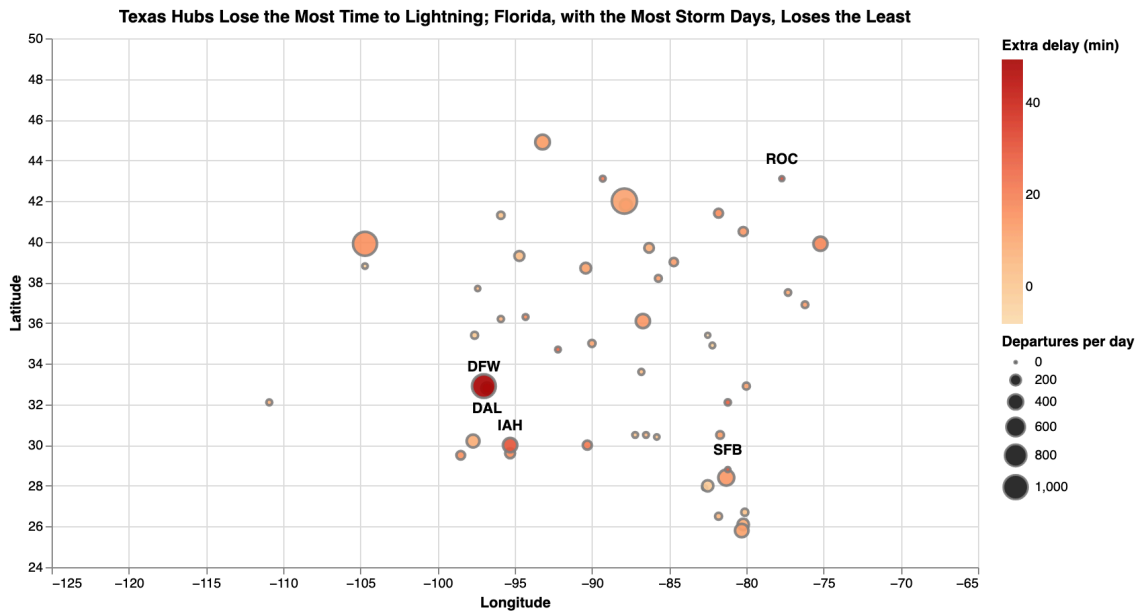
Most of what a storm costs is never recorded as weather delay. Question 4, "On Lightning-and-Hail Days, Late Aircraft Add More Delay than the Storm Itself", shows late-aircraft delay going up by 7.9 minutes per flight on the worst days, while weather delay goes up by 5.7. That delay also moves around the network. In Question 6, "Storms Elsewhere Delay Airports That Stay Dry", airports with no storm of their own

average 5.9 minutes of late-aircraft delay per flight on quiet days and 9.8 minutes on days with heavy lightning across the country. Question 3, "73% of Airport-Days with a BTS Weather Delay Had No Storm in the Airport Tile", fits with this. BTS can blame weather anywhere along a trip, while our data only looks at a small square around the departure airport, so the two often disagree, and neither one captures the whole cost on its own.



Question 4: "On Lightning-and-Hail Days, Late Aircraft Add More Delay than the Storm Itself"

You might expect the stormiest places to lose the most time, but Question 2, "Texas Hubs Lose the Most Time to Lightning; Florida, with the Most Storm Days, Loses the Least", shows the opposite. Florida airports have more lightning days than any other state but lose a median of only 5.5 extra minutes on those days, while DFW loses about 49. Our best guess is that airlines in Florida plan for afternoon storms because they happen so often, while a storm at a busy hub like DFW sets off a chain of late flights. Airlines differ too. Question 7, "SkyWest and JetBlue Absorb Storms Best; American and PSA Worst (Average 14.4 min)", compares each airline with itself at the same airports, and a storm day still costs SkyWest 8.7 extra minutes but PSA 22.5.



Question 2: "Texas Hubs Lose the Most Time to Lightning; Florida, with the Most Storm Days, Loses the Least"

A caveat: we only have one month of data (June 2026). Most airports had between 4 and 15 lightning days, and the DFW number rests on just 4. Storms just outside the 0.1° tile around an airport are also missed, so next we plan to add up the 3 x 3 block of tiles around each airport.

Streamlit application

Our Streamlit application has one page for each of our seven exploration questions, listed in the sidebar. Each page states its question, shows the chart that answers it, and gives a caption that says what the chart shows and how to use it. An expander below each chart holds the data behind it, and a footnote at the bottom cites BTS, NOAA NCEI and OurAirports. The application runs on Cloud Run in us-central1, reads the fused data from our Cloud Storage bucket, and is protected by Identity-Aware Proxy (IAP), so only invited Google accounts can open it:
https://storms-flight-delays-todd-652400839740.us-central1.run.app

In Milestone 4 we added an eighth page, "Ask your own question", under a "Vis Agent" heading in the sidebar. On this page, the user types a question about the data, and our Vis Agent answers it with a chart. The page shows the question, the dataset that the agent used, the chart, a description of the chart, and the code that made it. If the data cannot answer the question, the page shows the agent's explanation instead of a chart.

The four pages below have interactive elements. On the other three pages, hovering over a mark shows its numbers.

Deploy

Storms and Flight Delays

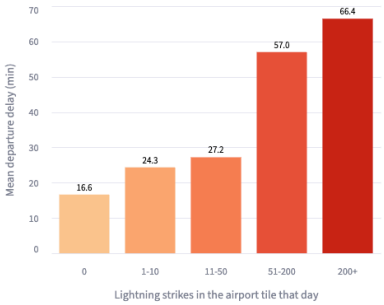
1. Lightning and delay
2. Storm-sensitive airports
3. BTS weather delay vs storms
4. Delay causes
5. Hail vs lightning
6. Delay that travels
7. Carriers in storms

Lightning and Departure Delay

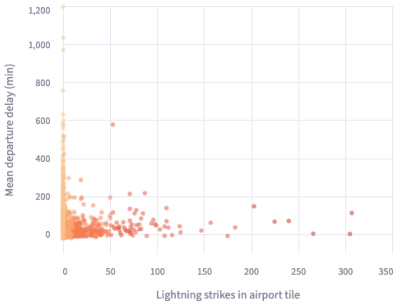
Question 1: How does the daily volume of lightning near an airport relate to that airport's mean departure delay?

An airport-day is one airport on one day in June 2026. For each airport-day, we count the lightning strikes in the 0.1° tile around the airport (about 11 x 9 km) and put the count into one of five levels.

Departure Delays Quadruple as Lightning at the Airport Increases



Every Airport-Day in the Selected Level



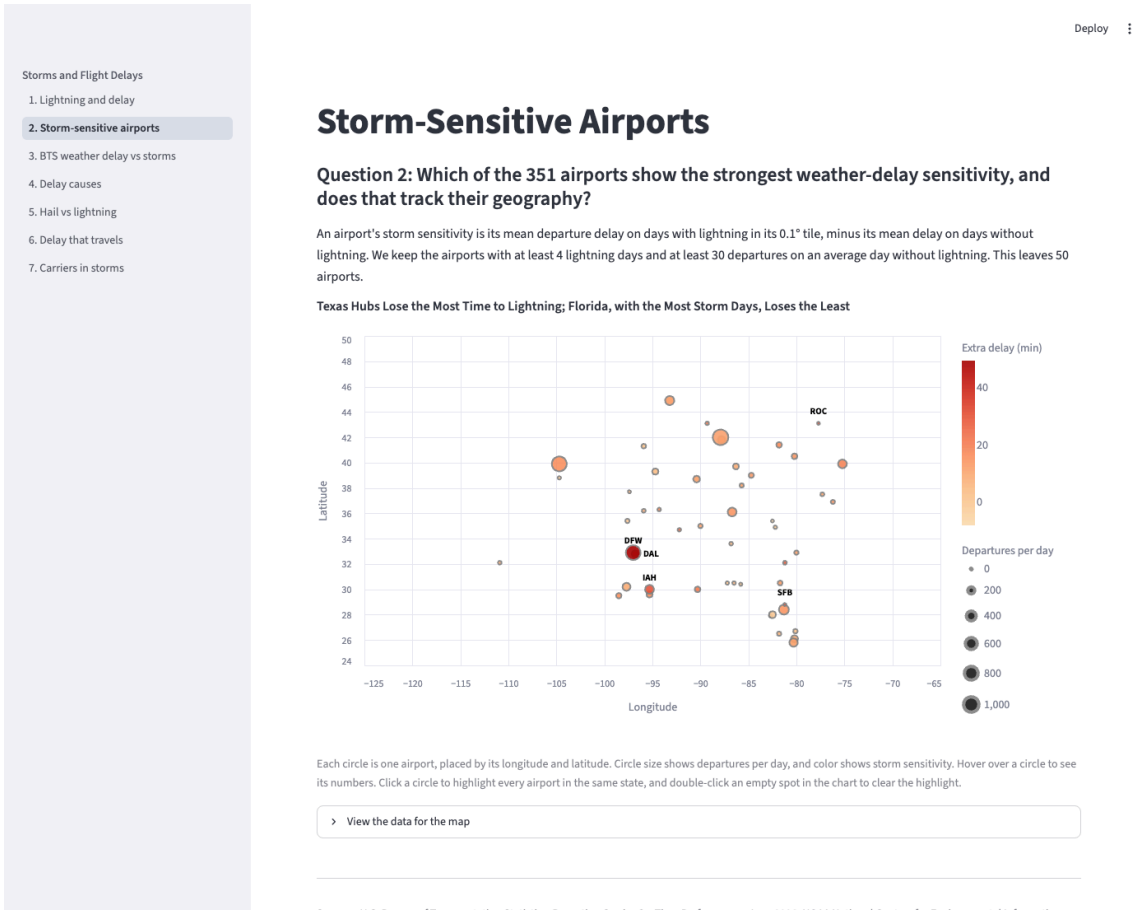
The bar chart shows the mean delay at each lightning level, and the scatter plot shows every airport-day. Hover over a bar to see how many airport-days it averages. Click a bar to show only that level in the scatter plot, and double-click an empty spot in a chart to show all levels again.

› View the data for the bar chart

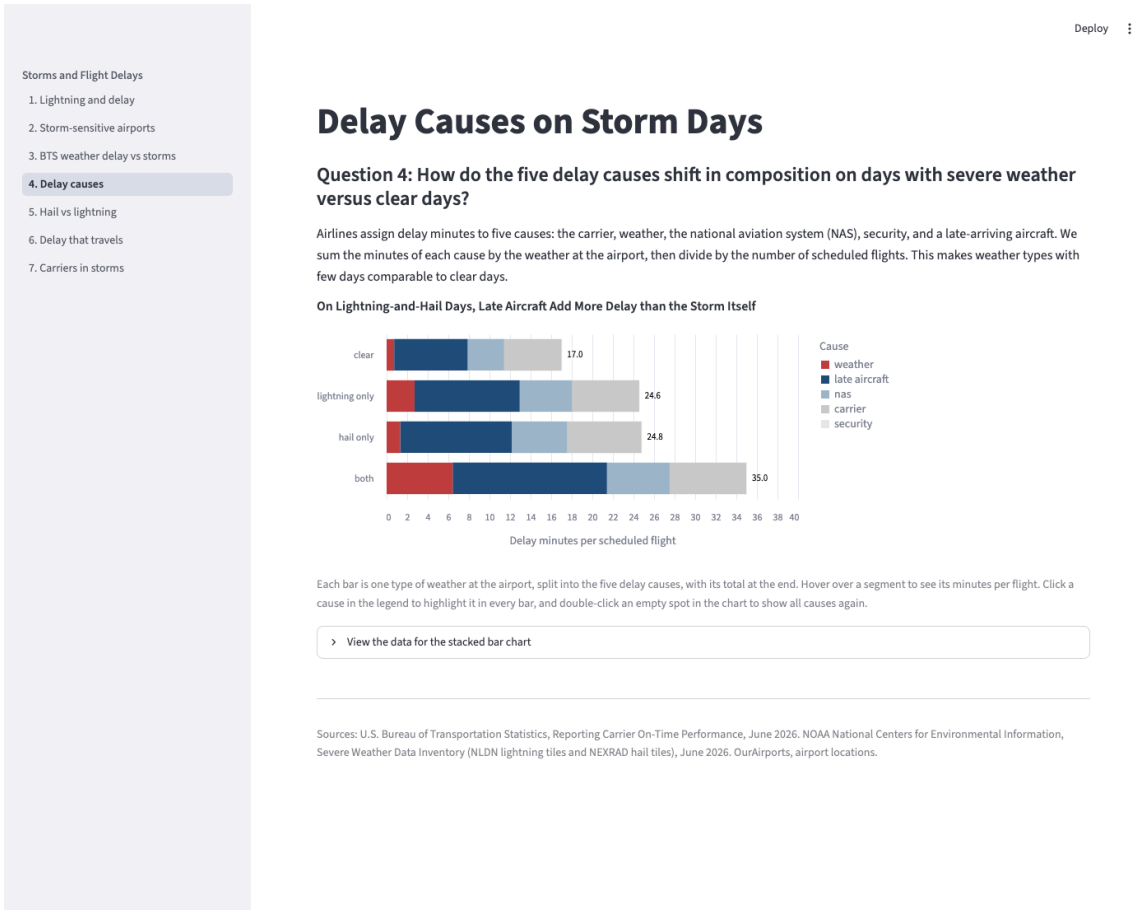
› View the data for the scatter plot

Sources: U.S. Bureau of Transportation Statistics, Reporting Carrier On-Time Performance, June 2026. NOAA National Centers for Environmental Information, Severe Weather Data Inventory (NLDN lightning tiles and NEXRAD hail tiles), June 2026. OurAirports, airport locations.

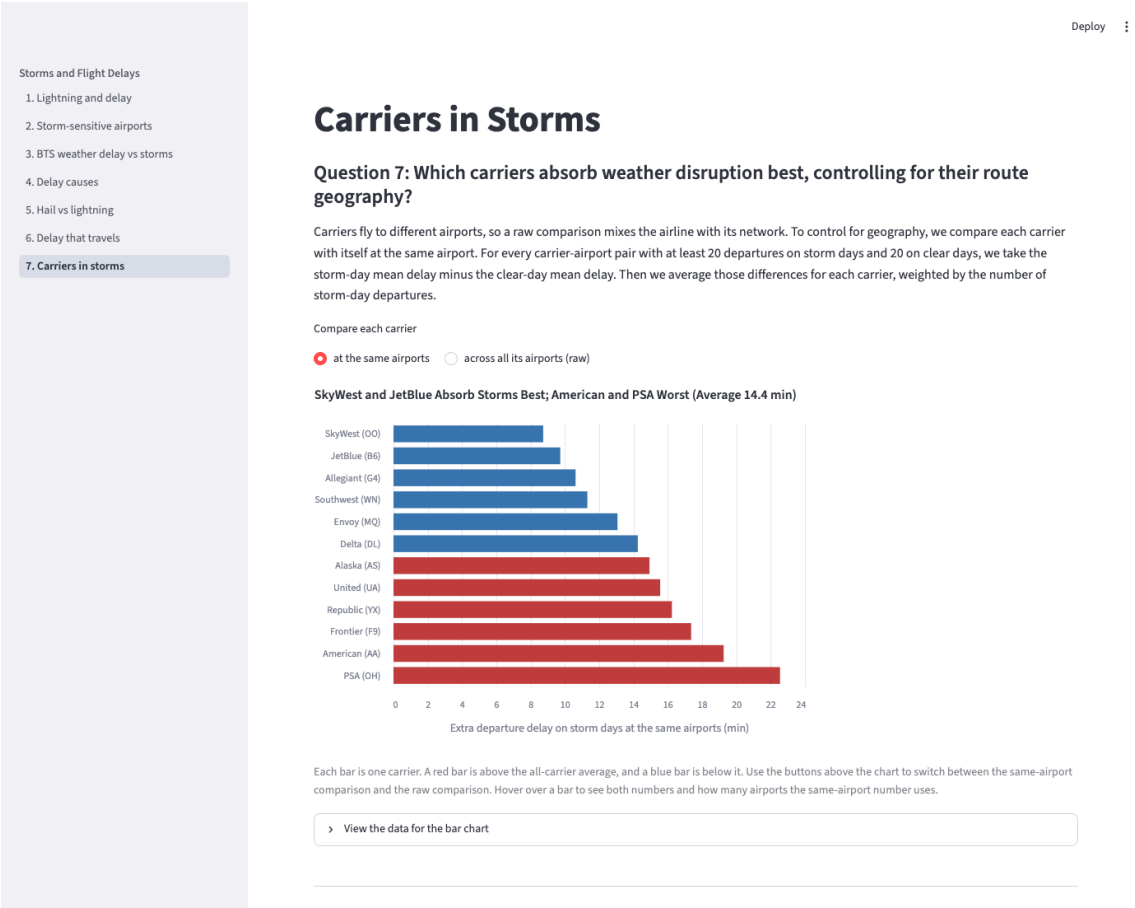
Question 1: click a bar to show only that lightning level in the scatter plot.



Question 2: click an airport to highlight every airport in the same state.



Question 4: click a delay cause in the legend to highlight it in every bar.

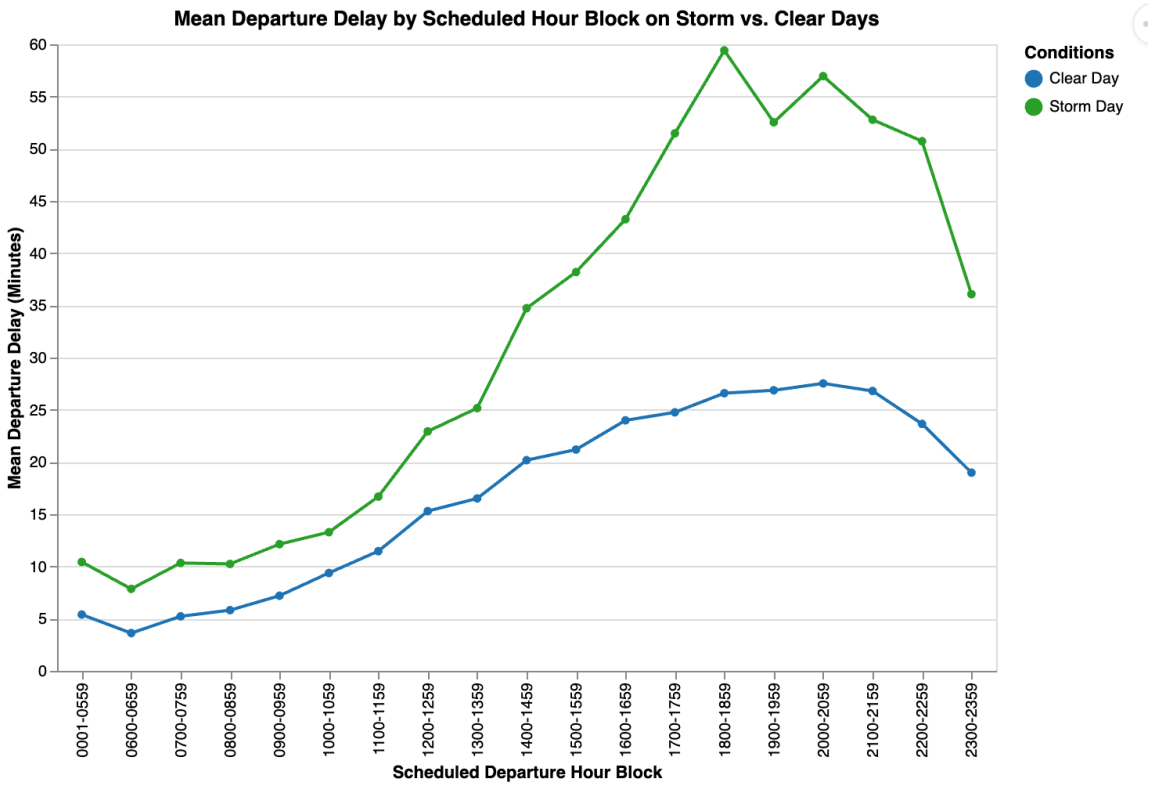


Question 7: the buttons switch between the same-airport comparison and the raw comparison.

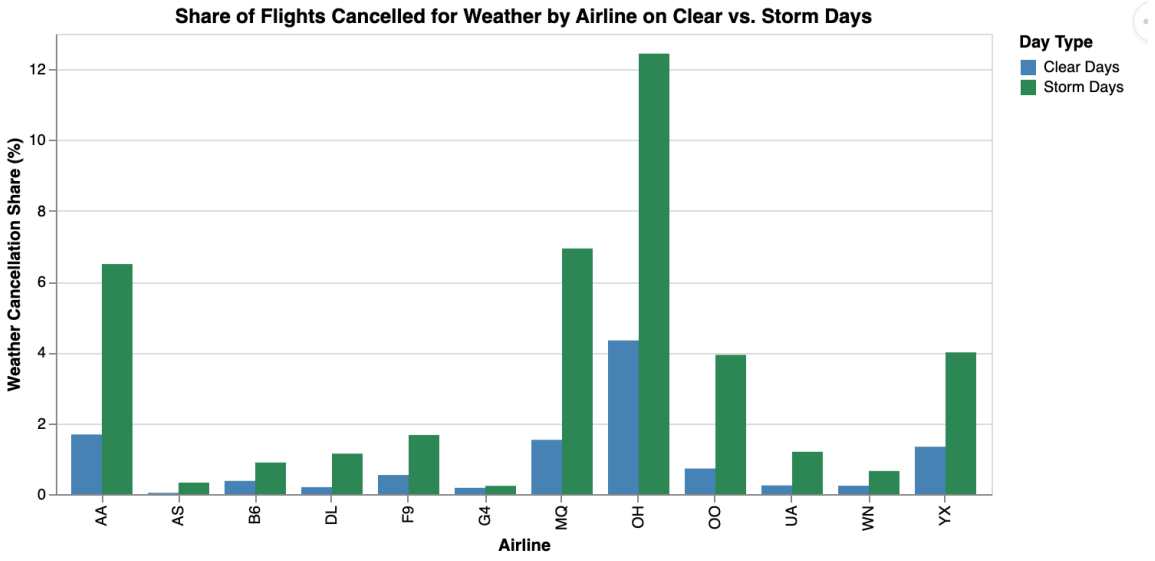
Vis Agent samples

The agent uses Gemini 3.5 Flash Lite and two datasets: airport_days (one row per airport per day) and flights (one row per flight). The samples below show a line chart, a grouped bar chart, a scatter plot, a faceted bar chart, and an answer for a question that the data cannot answer. Each sample shows the prompt, the dataset that the agent chose, and the chart that it made.

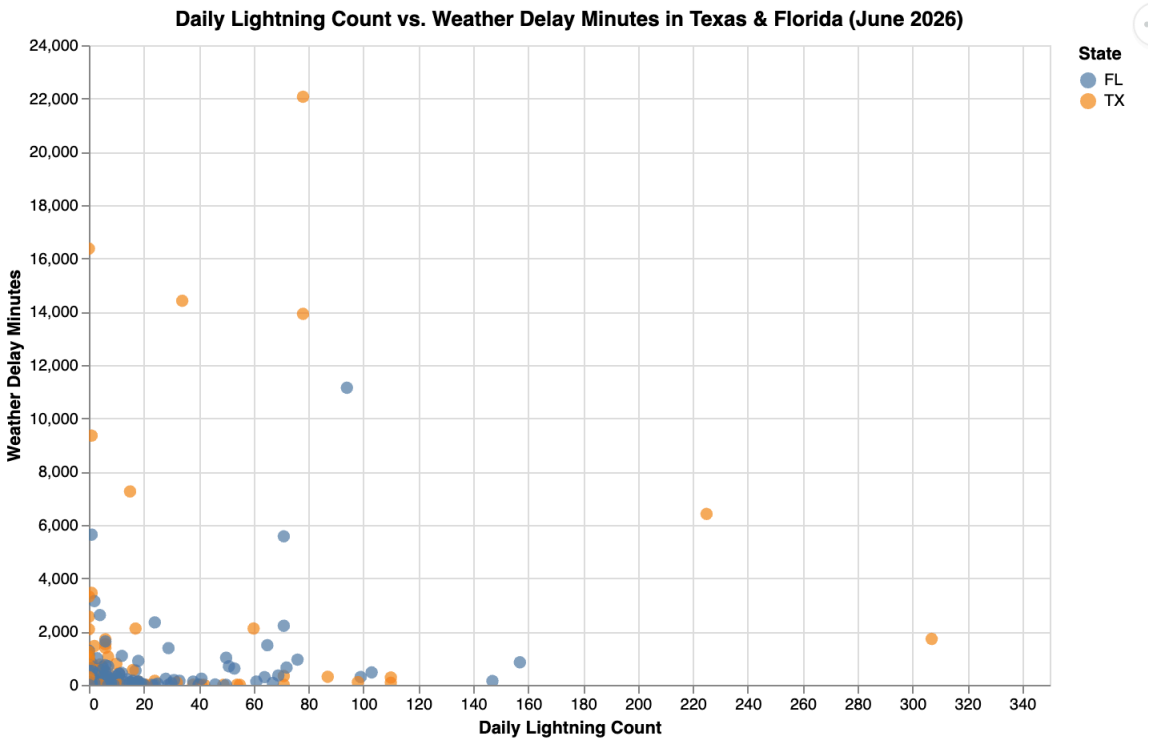
Prompt: "How does the mean departure delay change across scheduled departure hour blocks on storm days compared with clear days?" Dataset: flights.



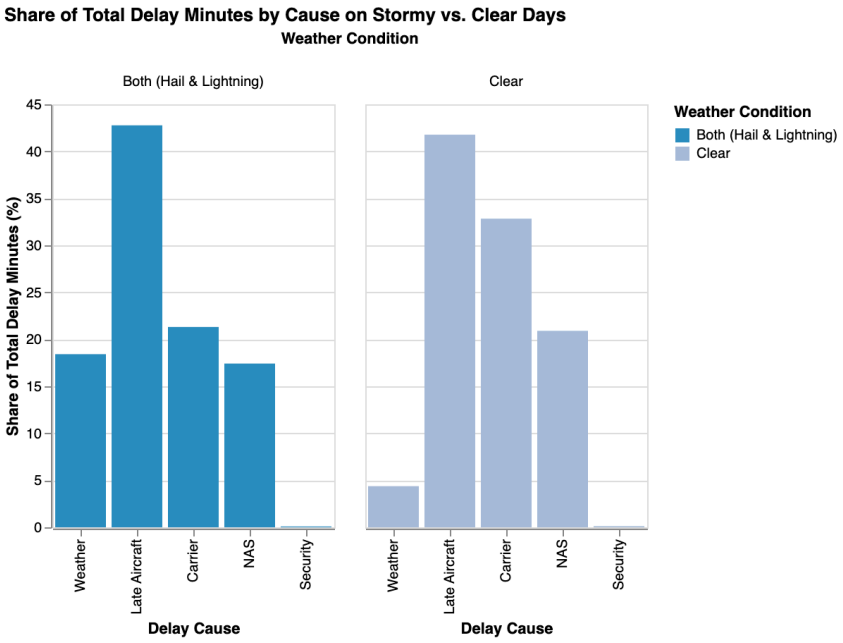
Prompt: "For each airline, compare the share of flights cancelled for weather (code B) on storm days with clear days." Dataset: flights.



Prompt: "For airports in Texas and Florida, how does the daily lightning count relate to the total weather delay minutes?" Dataset: airport_days.



Prompt: "Which delay cause takes a larger share of total delay minutes on days with both hail and lightning compared with clear days?" Dataset: airport_days.



Prompt: "How did flu activity change in Texas in 2025?" No dataset. The agent replied: "The available datasets contain information about US flight delays and storms in June 2026, but do not contain data

regarding flu activity or health statistics in Texas during 2025."

Evaluations

First user study

Summarize what users reported learning about the data through your visual analysis. Group recurring themes rather than listing every comment.

1. Storms cause delays, but they are not the main cause. Two users said that the other causes together delay more flights than storms do. One of them added that most late flights happen in any weather.
2. More lightning goes with longer delays. One user took this from the Question 1 page. Another user saw the same link but said that it is a correlation and not the main factor.
3. Texas airports lose more time to lightning than other airports. One user found this on the Question 2 page.

List the questions your users still had after exploring the application.

1. Can the map of storm-sensitive airports cover more of the US?
2. Why are some carriers worse than others at absorbing weather disruption?
3. How can carriers use these results when they plan flight schedules and routes?

Record the suggestions users offered for making the visualizations clearer or more useful.

1. Navigation and layout:
    a. The page menu in the top-left corner is small and easy to miss. Make the way to reach the other questions more visible.
    b. Put the answer directly under each question.
    c. The small gray text above the data tabs is hard to read. Make it larger or darker.

2. Charts and text:
    a. Give every chart a title.
    b. The Question 3 page (BTS weather delay vs. observed storms) needs more explanation for a general reader, because it does not say why the BTS attribution matters.
    c. The "Every Airport-Day in the Selected Level" chart could use a color scale to separate its points.
    d. The Question 5 chart is sparse. Another chart type could show the data better.
    e. Write equations as real equations, not in words.

Application enhancement

Identify which suggestion from the first user study you chose to implement and briefly explain why you selected it over the others.

- We chose the first question that users still had: "Can the map of storm-sensitive airports cover more of the US?" The Question 2 map showed only 50 airports, because we kept only airports with at least 4 lightning days and at least 30 departures per day. Most of the other suggestions were about titles, text size, and colors. Those changes do not change what a reader can learn from the data. This question asked for more data on the map, so it improves exploration.

Describe the changes you made to the Streamlit application to address that feedback.

- We replaced the two fixed filters with two sliders above the map: "Minimum lightning days" (1 to 10) and "Minimum departures per day" (1 to 100). Each slider has a help note that says what the reader gains and loses when they lower it. The text under the sliders now states the current filter values and how many of the 256 airports with lightning the map shows. If the reader moves a slider, the chart title changes to the number of airports, because the original title describes only the default view. The default values stay at 4 and 30, because they give the most reliable estimates. If the reader lowers both sliders, the map grows from 50 to more than 200 airports and covers the whole continental US.

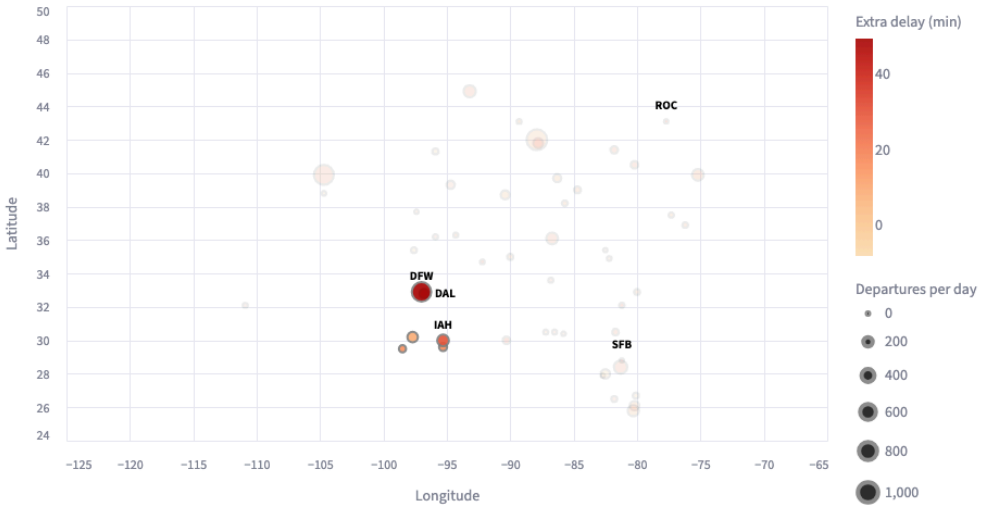
Include before and after screenshots of the affected page(s) so the changes are clearly visible.

Storm-Sensitive Airports

Question 2: Which of the 351 airports show the strongest weather-delay sensitivity, and does that track their geography?

An airport's storm sensitivity is its mean departure delay on days with lightning in its 0.1° tile, minus its mean delay on days without lightning. We keep the airports with at least 4 lightning days and at least 30 departures on an average day without lightning. This leaves 50 airports.

Texas Hubs Lose the Most Time to Lightning; Florida, with the Most Storm Days, Loses the Least



Each circle is one airport, placed by its longitude and latitude. Circle size shows departures per day, and color shows storm sensitivity. Hover over a circle to see its numbers. Click a circle to highlight every airport in the same state, and double-click an empty spot in the chart to clear the highlight.

› View the data for the map

Before: the Question 2 page from Milestone 3, with fixed filters and 50 airports.

Storm-Sensitive Airports

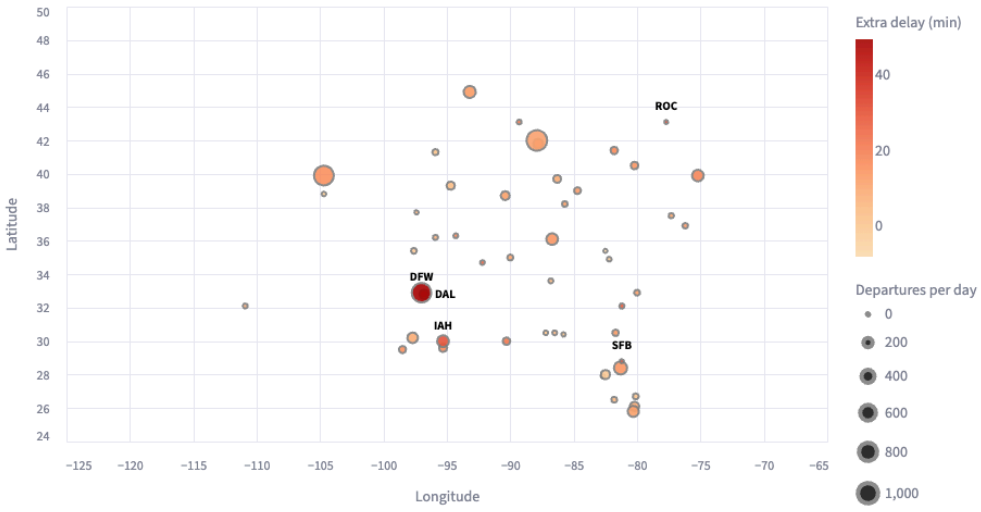
Question 2: Which of the 351 airports show the strongest weather-delay sensitivity, and does that track their geography?

Minimum lightning days 4

Minimum departures per day 30

An airport's storm sensitivity is its mean departure delay on days with lightning in its 0.1° tile, minus its mean delay on days without lightning. We keep the airports with at least 4 lightning days and at least 30 departures on an average day without lightning. This leaves 50 of the 256 airports that had lightning at least once. Lower the sliders to cover more of the US. The default values (4 and 30) keep only the most reliable estimates.

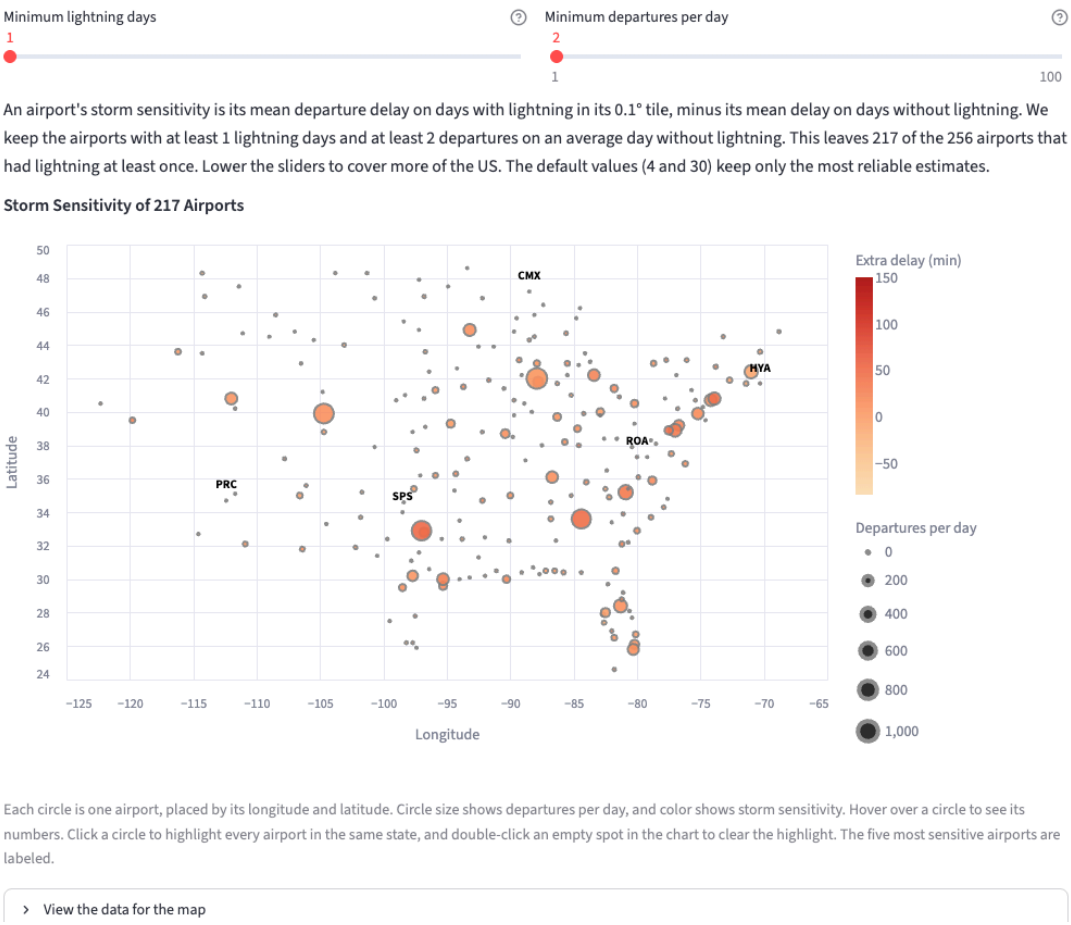
Texas Hubs Lose the Most Time to Lightning; Florida, with the Most Storm Days, Loses the Least



Each circle is one airport, placed by its longitude and latitude. Circle size shows departures per day, and color shows storm sensitivity. Hover over a circle to see its numbers. Click a circle to highlight every airport in the same state, and double-click an empty spot in the chart to clear the highlight. The five most sensitive airports are labeled.

› View the data for the map

After: the same page with the two new sliders at their default values (4 and 30).



After: with both sliders lowered, the map shows 217 airports across the US.

Agent evaluation study

Summarize the test questions and your observations of the agent's behavior across runs. Document three areas of improvement grounded in those observations.

- We asked the agent five questions, and each question needs a filter, an aggregation, or a comparison of two variables. We ran all five questions in order, then ran the full sequence a second time. The agent used Gemini 3.5 Flash Lite. The test questions were:
    1. For the 10 busiest airports, how does the mean departure delay on days with lightning compare with clear days?
    2. For each airline, compare the share of flights cancelled for weather (code B) on storm days with clear days.

3. How does the mean departure delay change across scheduled departure hour blocks on storm days compared with clear days?
4. For airports in Texas and Florida, how does the daily lightning count relate to the total weather delay minutes?
5. Which delay cause takes a larger share of total delay minutes on days with both hail and lightning compared with clear days?

Observations

Question 1 (airport_days):

- Both runs found the same 10 airports and the same values. Run 1 made a faceted bar chart with no tooltip, and its "Weather Condition" header repeats under each airport and the labels overlap. Run 2 made a grouped bar chart with a tooltip and a legend, which is easier to read. In both runs, 5 of the 10 airports (LAS, LAX, PHX, SEA, SFO) had no lightning days, so their lightning bar is missing. The chart does not tell the reader why.

Question 2 (flights):

- Both runs computed the same cancellation shares. In run 1, the first chart code raised an error, and the agent fixed it after test_chart returned the error. Run 1 made a grouped bar chart in blue and green. Run 2 made two side-by-side panels sorted from highest to lowest, which ranks the airlines but makes it harder to compare clear and storm days for one airline. Run 2 also used a light color for storm days, so the most important bars are the faintest.

Question 3 (flights)

- Both runs made almost the same line chart, with one line for storm days and one for clear days across the 19 hour blocks. Both charts have a tooltip and follow the color rule. This question gave the most consistent result.

Question 4 (airport_days)

- Both runs made the same scatter plot. The chart colors Texas orange, which breaks our rule to use blue and green first. Most points sit near zero on both axes, and a few days with more than 10,000 minutes of weather delay stretch the y-axis. The chart has no trend line or log scale, so the reader cannot see the relationship that the question asks about. The y-axis is total minutes per airport, so busy airports have more delay minutes only because they have more flights.

Question 5 (airport_days)

- Run 1 made a correct faceted bar chart: weather causes 18.4% of delay minutes on days with both hail and lightning and 4.4% on clear days. Run 2 made the same chart type, but its code gave the two groups the wrong names, so the chart shows the clear-day values under "Both Hail & Lightning". The reviewer approved both charts. Neither chart has a tooltip.

Across all runs. The agent chose the correct dataset in all 10 runs. Each run took between 6 and 27 seconds. The order of tool calls was the same each time. The agent called review_chart once per question, and the reviewer approved every chart, including the wrong one. Three of the 10 charts had no tooltip, although our rules ask for one (question 1 run 1, and question 5 in both runs). For a question about flu activity, the agent stopped after select_dataset and explained that our data does not cover that topic.

Areas of improvement

1. Review the data in the chart, not only the code. The reviewer reads the code and the agent's description, so it approved the question 5 run 2 chart with the wrong labels and the question 1 chart with missing bars. We will give the reviewer the aggregated table that the chart draws (we already extract it in test_chart) or an image of the rendered chart. Then the reviewer can compare the values with the question and catch swapped labels and empty groups.

2. Check the design rules in code, before the review. Tooltips were missing in three runs, and question 4 used orange. test_chart already renders the chart to its JSON specification, so it can also make sure that the chart has a tooltip encoding and uses only blue and green colors. If a rule fails, test_chart will return an error, and the model will fix the code the same way that it fixed the question 2 error.

3. Make the chart choice more consistent and fit the measure to the question. Questions 1 and 2 changed between a faceted chart and a grouped chart from one run to the next, and the grouped charts were easier to compare. We will add rules to design_chart: use a grouped bar chart when the question compares two conditions, use a darker color for the storm condition, say in the description when a group has no data, and use a rate per flight or a log scale when a measure is skewed, as in question 4. We will also lower the temperature from 0.1 to 0, so the same question gives the same chart more often.

Second user study (extra credit)

List the questions users asked your Vis Agent.
Summarize what they learned about the data. Group recurring themes rather than listing every comment.
Record the suggestions users offered for making the agent more functional and useful.